

Face Detection Using U-Net-based Skin Filtering and ML -based Triplet Scoring

Yu-Wei Hsu, National Taiwan University

GitHub Repository: <https://github.com/Yaweihsu0201/Rule-Based-Face-Detection>

May 2026

Abstract

This project implements and improves existing face feature detection methods[1] for locating the eye and mouth positions from an input image. The original method uses skin filtering, ellipse matching, eye maps, mouth maps, and hand-crafted geometric rules to select the best eye-eye-mouth triplet. However, traditional skin filtering is sensitive to illumination, background noise, and skin color variations, while rule-based triplet scoring relies on manually designed weights that may not generalize well across different images. To address these limitations, this project introduces a U-Net-based skin filtering model to generate a more robust skin mask for ellipse matching. Based on the detected face ellipse, eye and mouth candidates are extracted from the eye map and mouth map. Candidate triplets are then constructed by combining two eye candidates and one mouth candidate. Instead of using only hand-crafted scoring rules, this project formulates triplet selection as a machine learning-based scoring problem. For each triplet, ellipse-relative geometric features are extracted, including normalized positions in the ellipse coordinate system, normalized distances, angular relationships, ellipse shape and orientation, and eye/mouth map responses. A classifier is trained to predict whether a candidate triplet forms a valid facial structure. Experimental observations show that the proposed improvements increase the flexibility of triplet selection and reduce the dependency on manually tuned geometric scoring rules.

1. Introduction

Face detection is an important task in the field of Computer Vision, aiming to find the facial region of an input image and locate the facial features such as eyes and mouth. Such technologies can be applied in scenarios such as face recognition, emotion analysis, semantic understanding and human-machine interaction. However, robust face detection remains challenging in real-world situations because of illumination changes, background noise, skin color variations, facial rotation, and inaccurate candidate points.

This project is based on a previous face detection method proposed in an earlier thesis. The original method first applied skin filtering on an input image to extract possible skin regions, then performs PCA on each region to estimate each region as an approximate ellipse. Based on the detected face ellipse, eye maps and mouth maps are computed to generate candidate eye and mouth points. Finally, two eye candidates and one mouth candidate are combined into an eye-eye-mouth triplet, and a hand-crafted geometric scoring function is used to select the most plausible facial structure. Although the original method utilized meaningful facial

geometry, it still has several limitations. The original method relies on classical skin filtering and fixed-weight triplet scoring, which may be inaccurate under illumination changes and background noise.

To address those limitations, this project improves two main components of the original method. First, a U-Net model is introduced for skin filtering, converting the input RGB image into a skin mask. Second, a ML-based scoring system is introduced for triplet selection. For each candidate eye-eye-mouth triplet, ellipse-relative geometric features are extracted, including normalized distances, normalized positions in the ellipse coordinate system, angular relationships, ellipse shape, orientation, and eye/mouth map response values. A classifier is then trained to predict whether a candidate triplet forms a valid facial structure.

Overall, this project does not propose a completely new face detection pipeline from scratch. Instead, we implement, analyze and extend an existing ellipse-based feature detection method. The main contributions are the integration of U-Net-based skin filtering and the use of machine learning-based triplet scoring to reduce the dependency on manually tuned geometric scoring rules.

2. System overview

The pipeline of this project is shown in the figure below. First, the system applies skin filtering on the input image using a pre-trained U-Net model and extract a skin mask. Based on the original method, the skin mask is segmented into clusters, and each cluster is estimated as an ellipse with parameters such as centers, major axis and minor axis. Then, the system computes the eye-map and mouth-map, selecting most possible position of eyes and mouths on each ellipse. Finally, the system computes every combination of eye-mouth triplet through a machine learning-based scoring method to select the most plausible face detection.

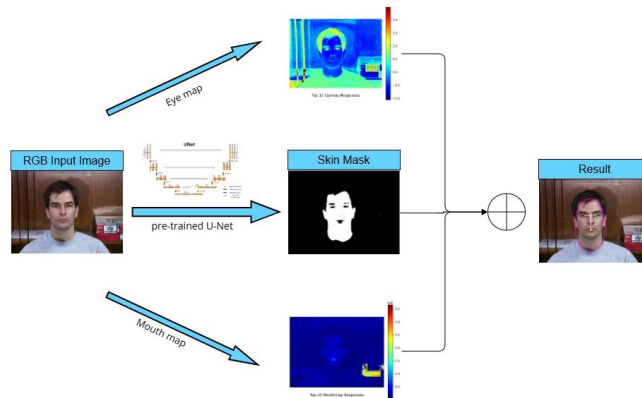


Figure 1. System pipeline

3. Original Method

The original Method proposed in earlier thesis detects the position of eyes and mouth based classical image filtering and geometry rules. The pipeline consists of four stages: skin filtering, ellipse matching, eye/mouth map and geometry-based triplets' detection. First, the system extracts possible skin-like regions. Then, for each region, the system estimates the approximate ellipse for each region. Second, the system computes the eyemap and mouthmap of the input image and selects the several candidates of eye-candidates and mouth-candidates. Finally, the system find the most plausible eye-mouth triplet inside the ellipse and determine as a face if it is found.

3.1 Skin Filtering

In the original method, the skin filter filters out pixels that don't look like skin, so the system only needs to process information inside the ellipse of each skin-like region.

Classical skin-filtering usually evaluates pixels based on the color space information, such as RGB, YCbCr or HSV, to produce a skin mask. The pixels inside the skin mask are highly possible to be a skin pixel, thus can be the input of next stage.

3.2 Ellipse Matching

4. Proposed Improvements

This project extends the original ellipse-based face feature detection method by improving two major components: skin filtering and triplet scoring. Instead of relying only on traditional color-based skin filtering, a U-Net model is used to generate a more robust skin mask. In addition, the original hand-crafted triplet scoring function is replaced with a machine learning-based scoring model that learns from labeled triplet data.

4.1 U-Net based skin-filtering

4.1.1 Motivation

Traditional skin filtering is sensitive to lighting conditions. The algorithm used in original method is SVM, which is sensitive to shadows, different races and lighting conditions. Furthermore, fixed color thresholds may fail under different skin tones or backgrounds. Therefore, the precision and recall of this method is usually low. Poor skin masks affect ellipse matching and later facial feature detection, so it is essential to address this problem.



Figure 2. Error due to shadow



Figure 3. Error due to different skin tones

4.1.2 Architecture

The skin filtering model used in this project is a simplified U-Net architecture for image segmentation. The model takes a 3-channel RGB image as input and outputs a single-channel skin mask. It follows an encoder-decoder structure with skip connections.

In the encoder path, the input image is passed through several convolution blocks. Each convolution block contains two convolutional layers, followed by batch-normalization and ReLU activation. After each block, max pooling is applied to reduce the spatial resolution. As the network goes deeper, the number of feature channels increases from $64 \rightarrow 128 \rightarrow 256 \rightarrow 512$, allowing the model to learn increasingly abstract and high-level features.

In the decoder path, transposed convolution is used to up-sample the feature maps and gradually recover the original spatial resolution. At each decoding stage, the up-sampled feature map is concatenated with the corresponding encoder feature map through a skip connection. These skip connections help preserve low-level spatial details, which is important for producing accurate segmentation boundaries.

Finally, a 1×1 convolution is used to reduce the feature channels to one output channel, producing the final skin segmentation mask.

4.1.3 Role in the pipeline

An RGB image and a pretrained weight will be fed as an input into the U-Net model. The model will output a binary image as skin mask, where 1 represents skin-like pixel and 0 otherwise.

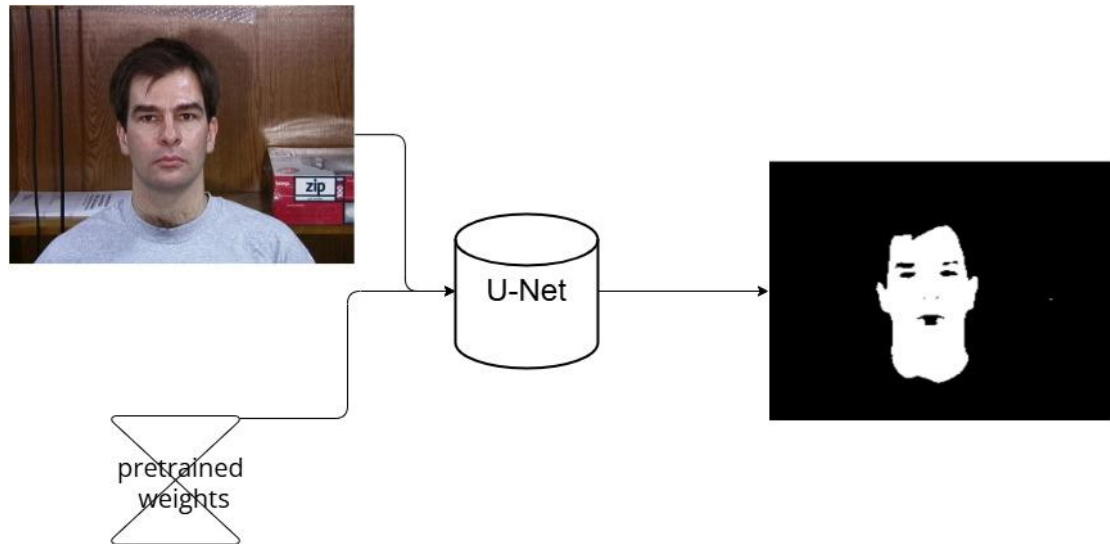


Figure 4. U-Net skin filtering

4.2 Machine Learning-based Triplet Scoring

4.2.1 Limitation of Rule-based Scoring

The original method uses manually designed weights and a rule-based method to select the most plausible triplets. Because thresholds and penalties may not generalize well, different images may require different scoring behavior.



Figure 5. False results of triplets

4.2.2 Triplet Classification Formulation

Input:

A candidate triplet consisting of left eye, right eye, and mouth.

Output:

Probability that the triplet is a valid facial structure.

Label:

1 = valid triplet

0 = invalid triplet

4.2.3 Feature Categories

1. Normalized distances between eyes and mouth
2. Ellipse-normalized positions of eyes and mouth
3. Distances from ellipse center
4. Angular relationships between long-axis and eyes-mouth
5. Eye map and mouth map response values

4.2.4 Model Training and Triplet Selection

Candidate triplets are generated from each image. Then, Triplets are manually labeled as valid or invalid. A classifier (in this project we used random forest) is trained to predict triplet validity. During testing, the triplet with the highest predicted score is selected.

After all possible candidate triplets are generated, the features are extracted for each triplet. The trained model predicts a validity score, which all triplets will be sorted by. Then, the top-ranked triplet is selected as the result.

4.3 Summary of Improvements

Component	Original Method	Improved Method
Skin filtering	Color/pixel-based filtering	U-Net-based skin filtering
Triplet scoring	Hand-crafted geometric score	Machine learning-based score

5. Experiments

5.1 Experimental Objective

The goal of the experiments is to evaluate whether the proposed improvements can enhance the robustness of the original face feature detection method. Specifically, the experiments focus on two parts: the quality of U-Net-based skin filtering and the effectiveness of machine learning-based triplet scoring for selecting the correct eye-eye-mouth configuration.

5.2 Dataset Preparation

For the training of U-Net, this project used Face and Skin Detection (FSD) Database. It contains 4,000 color images designed to support research in skin segmentation, used to

develop models for classifying and quantifying skin pixels, as well as detecting face regions.

For triplet scoring, each triplet consists of two eye candidates and one mouth candidate. The triplets generated were manually labeled as valid or invalid:

label = 1: valid eye-eye-mouth triplet

label = 0: invalid triplet

label = -1: unlabeled, not used for training

5.3 Feature Extraction

For each candidate triplet, features were extracted based on the detected face ellipse. The features include normalized distances, ellipse-relative coordinates, distances from the ellipse center, angular relationships, ellipse shape and orientation, and eye/mouth map response values. These features were used as the input to the machine learning classifier.

6. Results

6.1 Skin Filtering

The U-Net-based skin filtering model was able to extract major skin regions from the input images. The resulting skin masks provided the basis for ellipse matching and reduced the search region for facial feature detection.

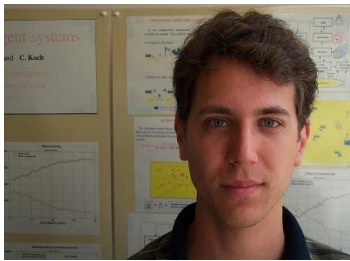
Input Image	Original	Improved
		
		

Figure 6. Improvements in skin filtering

6.2 Triplet Scoring Results

The machine learning-based scorer ranks candidate triplets based on their ellipse-relative geometric features and map response values. In many cases, the correct eye-eye-mouth configuration receives a higher score than geometrically unreasonable candidates.



Figure 7. Examples of triplet scorings

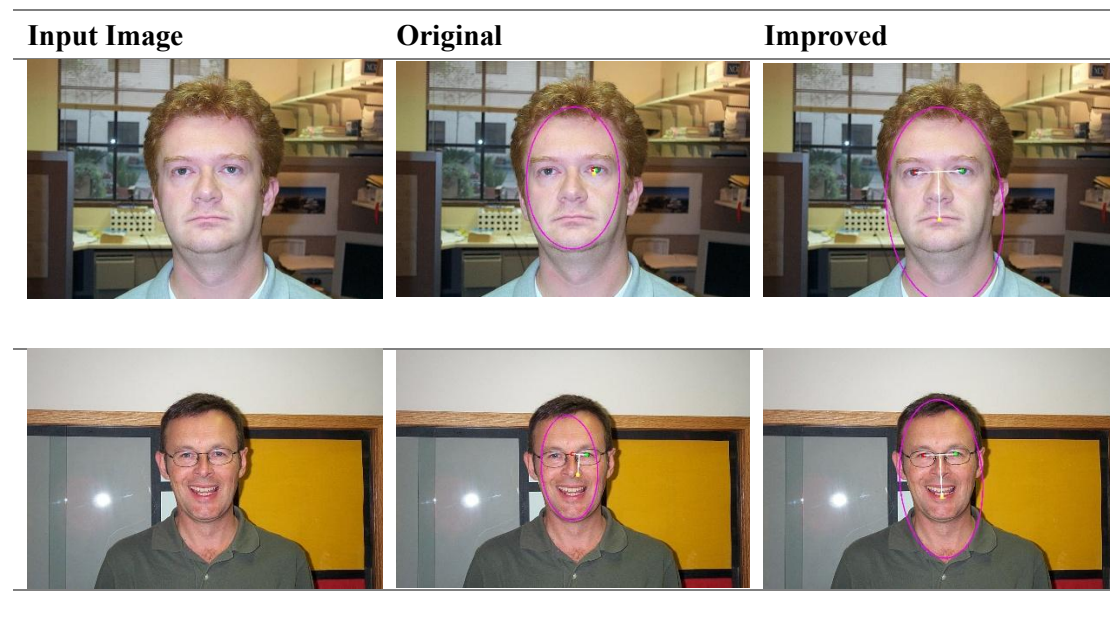


Figure 8. Improvements on triplet scroing

6.3 Comparison with original method

In this section, we compare the methods and results of the original to those of the improved on the same dataset.

Methods:

Method	Scoring Strategy	Strength	Limitation
Original method	Pixel-based skin filter, Hand-crafted geometric score	Simple and interpretable	Requires manually tuned weights
Improved method	DL-based skin filter, ML-based triplet score	Learn from labeled data	Depends on training data quality

Results:

Method	Precision	Recall	F1 Score
Original method	31/39=79.49%	41/72=56.94%	66.35%
Improved method	64/70=91.43%	64/74=86.49%	88.89%

Examples:



7. Discussion

From the qualitative results, our method performs well on frontal faces with clear facial structures. The ellipse-based face candidate generation and triplet-based scoring help reduce false positives in simple backgrounds. However, the method becomes less stable when the face is partially occluded or located in a complex background.

7.1 Failure cases and limitations

1. Skin-based face detection:

Our method relies heavily on skin-filtering results. Although we achieved high precision on skin-filtering, the results are still false during: (1) extreme light conditions (2) non-frontal face (3) disturbance caused by skin-like objects (hands, neck...). Because skin-filtering is the first step in our pipeline, failure in this step leads to complete false output. Below we show some of the failed cases:



Figure 9. Failed cases

2. Eyemap and Mouthmap:

The eyemap and mouthmap determines the candidates for triplet scoring, so they are critically important for the results. There are still limitations on eyemap and mouthmap in some cases: (1) Eyebrows are sometimes misclassified as eyes because they share similar visual characteristics with eye regions, such as dark color, horizontal shape, and symmetric placement. In low-resolution or low-contrast images, the actual eyes may be less salient than the eyebrows, causing the detector to assign higher scores to eyebrow regions. (2) The mouth point is sometimes detected between the nose and the actual mouth because this region also contains strong dark edges and shadows, such as nostrils, the philtrum, and the shadow under the nose. Those limitations indicates that relying only on handcrafted geometric and intensity-based features is not sufficiently robust.

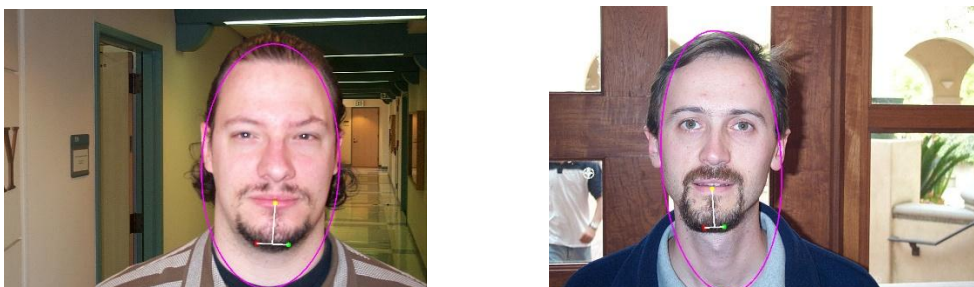


Figure 10. Failed cases

8. Future Work

a. Improve robustness for facial feature verification

Since the current method relies on frontal-face geometric assumptions, its performance may decrease when the face is rotated or partially turned away from the camera.

In future work, the system can be improved to handle non-frontal faces and large pose variations. Also, a more robust facial feature verification step can be added to reduce false detections. For example, additional constraints based on texture, intensity distribution, and local shape can be used to distinguish eyes from eyebrows and mouths from nose shadows.

b. Learning-based classification

Another possible improvement is to combine the rule-based pipeline with a learning-based classifier. The current method can be used to generate candidate face regions, and a small CNN can be trained to verify whether each candidate is a true face or a false positive.

c. Dataset diversity

More diverse test images can be collected to evaluate the method more comprehensively. The dataset should include different lighting conditions, face sizes, facial poses, occlusions, and complex backgrounds. Also, the detection pipeline can be extended with a multi-scale strategy so that faces of different sizes can be detected more reliably. This would be especially useful for small faces or images with multiple people.

d. Tune scoring weights automatically

The scoring weights and thresholds can be optimized automatically using a validation set. Instead of manually adjusting parameters, grid search or other optimization methods can be applied to find more stable parameter settings.

9. References

[1] 廖科傑, “運用輪廓色彩和五官的人臉偵測技術,” 國立臺灣大學電信工程學研究所碩士論文, 2010.

[2] O. Ronneberger, P. Fischer, and T. Brox, “U-Net: Convolutional Networks for Biomedical Image Segmentation,” MICCAI, 2015.

[3] C. M. Bishop, Pattern Recognition and Machine Learning, Springer, 2006.